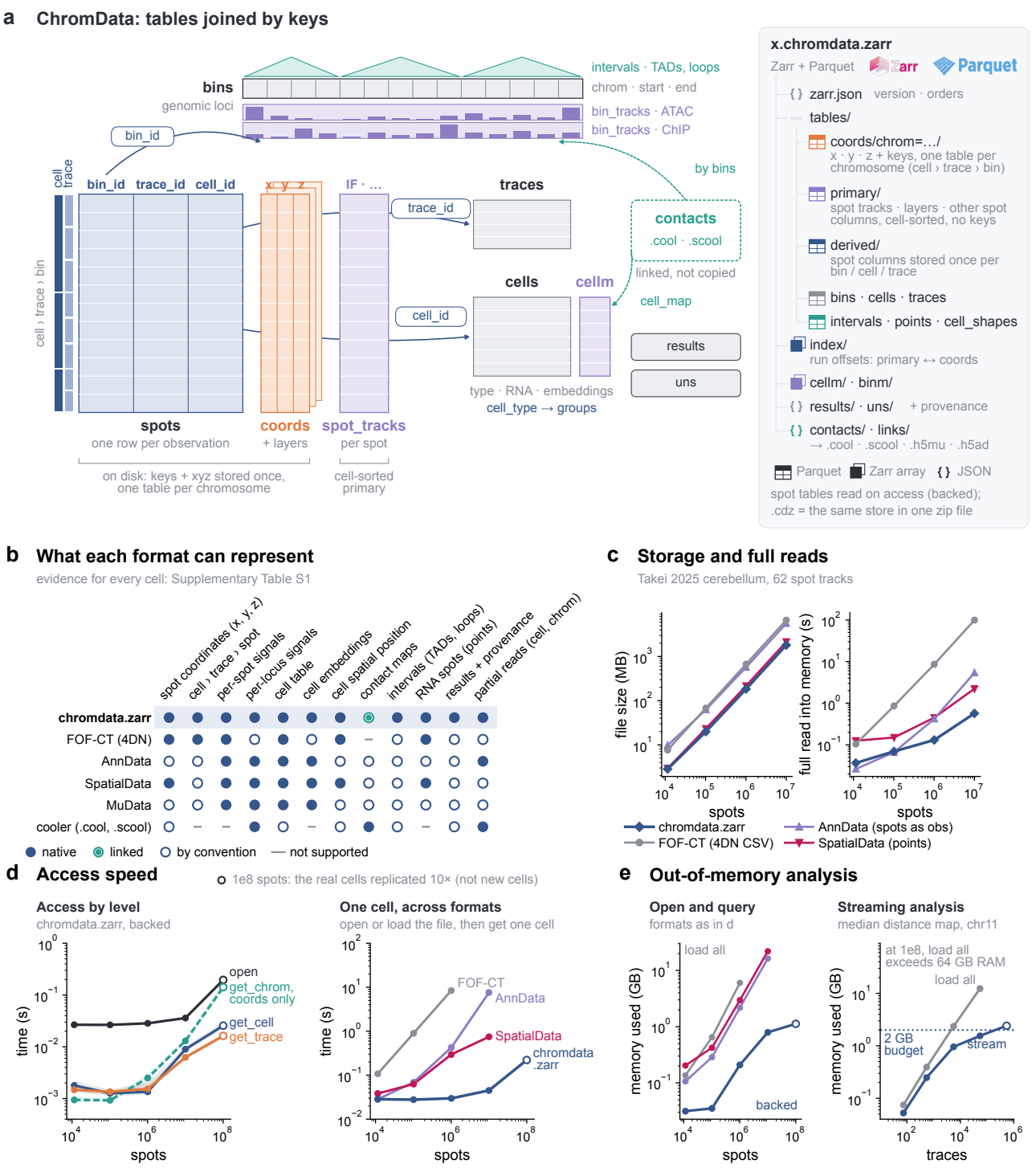

a
ChromData: tables joined by keys
bins
genomic loci
intervals · TADs, loops
chrom · start · end
bin_tracks · ATAC
bin_tracks · ChIP
bin_id
by bins
cell
trace
bin_id
trace_id
cell_id
x y z
IF · …
traces
trace_id
contacts
.cool · .scool
linked, not copied
cells
cellm
cell_map
cell_id
results
uns
cell › trace › bin
type · RNA · embeddings
cell_type → groups
spots
one row per observation
coords
+ layers
spot_tracks
per spot
on disk: keys + xyz stored once,
one table per chromosome
cell-sorted
primary
x.chromdata.zarr
Zarr + Parquet
zarr
Parquet
zarr.json
version · orders
tables/
coords/chrom=…/
x · y · z + keys, one table per chromosome (cell › trace › bin)
primary/
spot tracks · layers · other spot columns, cell-sorted, no keys
derived/
spot columns stored once per bin / cell / trace
bins · cells · traces
intervals · points · cell_shapes
index/
run offsets: primary ↔ coords
cellm/ · binm/
results/ · uns/
+ provenance
contacts/ · links/
→ .cool · .scool · .h5mu · .h5ad
Parquet
Zarr array
JSON
spot tables read on access (backed);
.cdz = the same store in one zip file
b
What each format can represent
evidence for every cell: Supplementary Table S1
spot coordinates (x, y, z)
cell › trace › spot
per-spot signals
per-locus signals
cell table
cell embeddings
cell spatial position
contact maps
intervals (TADs, loops)
RNA spots (points)
results + provenance
partial reads (cell, chrom)
chromdata.zarr
FOF-CT (4DN)
AnnData
SpatialData
MuData
cooler (.cool, .scool)
native
linked
by convention
not supported
c
Storage and full reads
Takei 2025 cerebellum, 62 spot tracks
file size (MB)
full read into memory (s)
spots
spots
chromdata.zarr
FOF-CT (4DN CSV)
AnnData (spots as obs)
SpatialData (points)
d
Access speed
1e8 spots: the real cells replicated 10× (not new cells)
Access by level
chromdata.zarr, backed
time (s)
open
get_chrom, coords only
get_cell
get_trace
spots
One cell, across formats
open or load the file, then get one cell
time (s)
FOF-CT
AnnData
SpatialData
chromdata .zarr
spots
e
Out-of-memory analysis
Open and query
formats as in d
memory used (GB)
load all
backed
spots
Streaming analysis
median distance map, chr11
memory used (GB)
at 1e8, load all exceeds 64 GB RAM
load all
2 GB budget
stream
traces